

# DNA REPLICATION

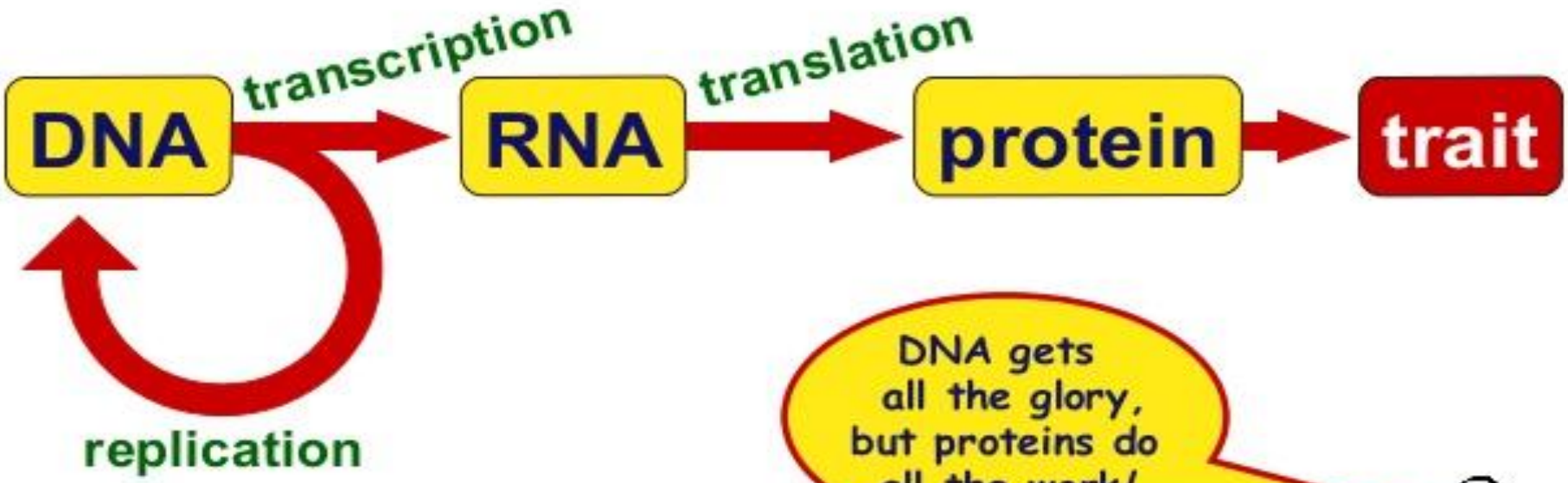
# Flow of information: from DNA (genes) to proteins

- **Unidirectional expression of genes**



## The "Central Dogma"

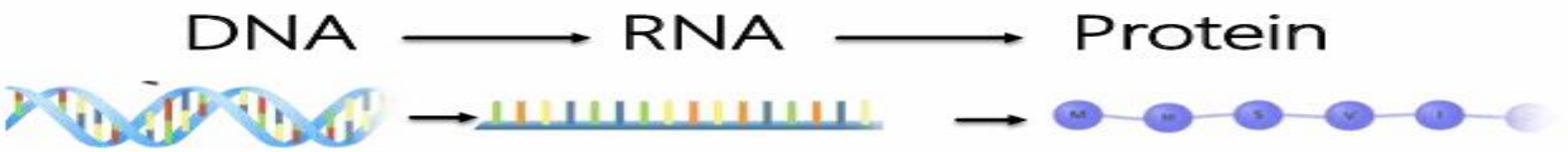
- Flow of genetic information in a cell
  - How do we move information from DNA to proteins?

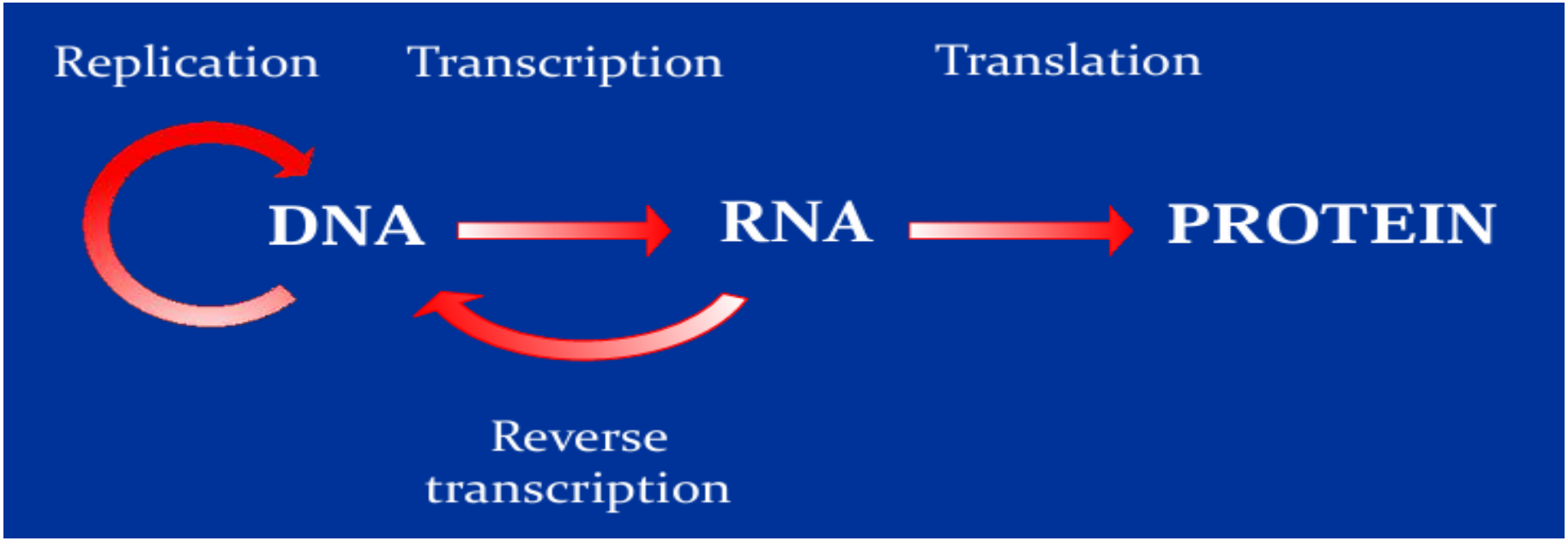




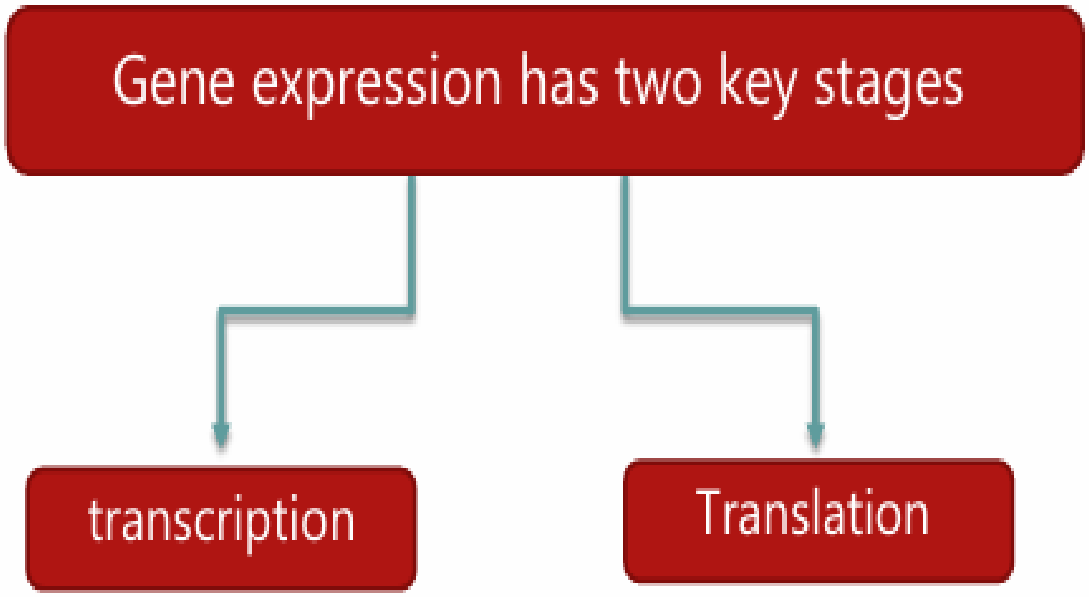

# Central dogma

It's explain the flow of genetic information, from DNA to RNA , to make a functional product, which is A protein .



The process by which the DNA instructions are converted into the functional product is called **gene expression** .



# Transcription and Translation

In transcription, the information in the DNA of every cell is converted into small, portable RNA messages.

During translation, these messages travel from where the DNA is in the cell nucleus to the ribosomes in cytoplasm where they are 'read' to make specific proteins.

# DNA Replication

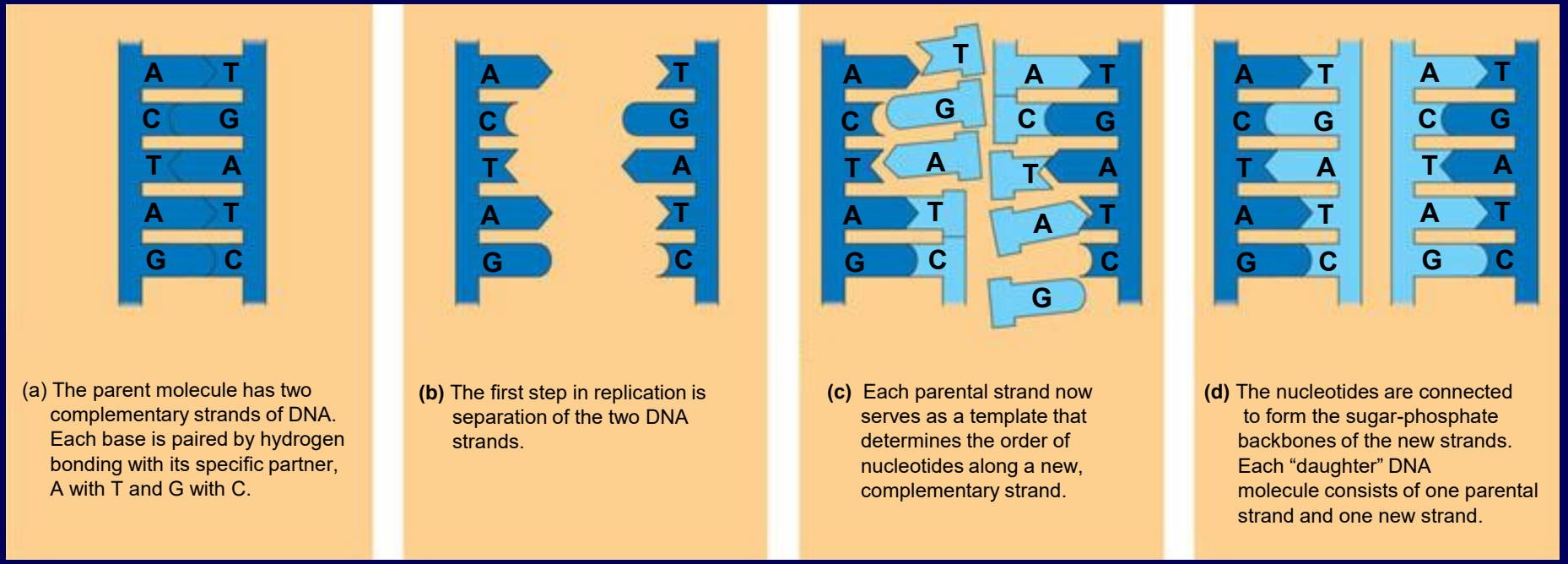
- DNA replication is a biological process that occurs in all living organisms and copies their exact DNA. It is the basis for biological inheritance.
- Replication is the process of synthesis of daughter DNA from parental DNA by the enzyme DNA Polymerase.
- $(dNMP)_n + dNTP \rightarrow (dNMP)_{n+1} + PPi$

  DNA → Lengthened DNA

- Each cell must replicate its DNA before cell division to transfer genetic information to daughter cells.

# DNA replication

- The parent molecule unwinds, and two new daughter strands are built based on base-pairing rules




(a) The parent molecule has two complementary strands of DNA. Each base is paired by hydrogen bonding with its specific partner, A with T and G with C.

**(b)** The first step in replication is separation of the two DNA strands.

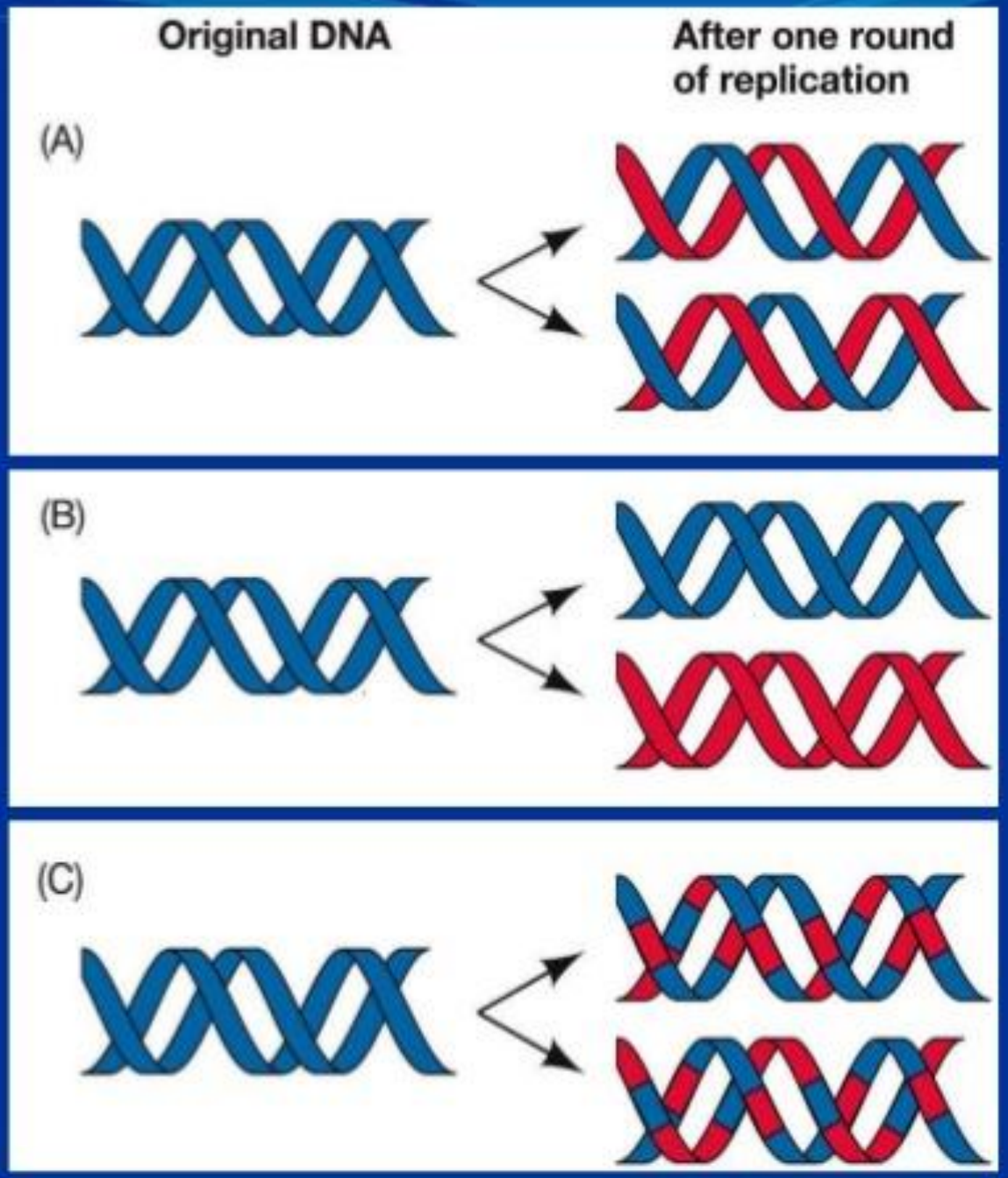
**(c)** Each parental strand now serves as a template that determines the order of nucleotides along a new, complementary strand.

**(d)** The nucleotides are connected to form the sugar-phosphate backbones of the new strands. Each "daughter" DNA molecule consists of one parental strand and one new strand.

*Semiconservative replication*

*Conservative replication*

*Dispersive replication*

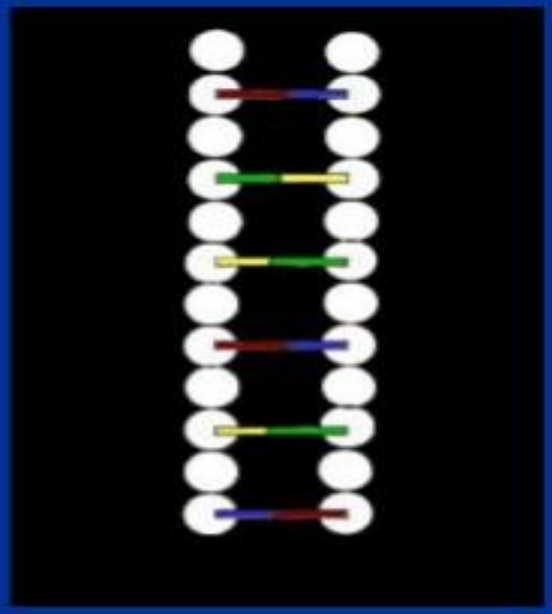


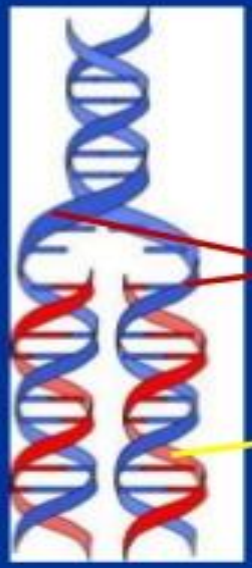
# DNA REPLICATION IS SEMI CONSERVATIVE

- Each replicated duplex daughter DNA molecule contains **one parent strand** and **one newly synthesized strand.**
- The base pairing rule is always maintained.



*Each parent strand serves as a template for a new strand and the two new DNA strands each have one old and one new strand*

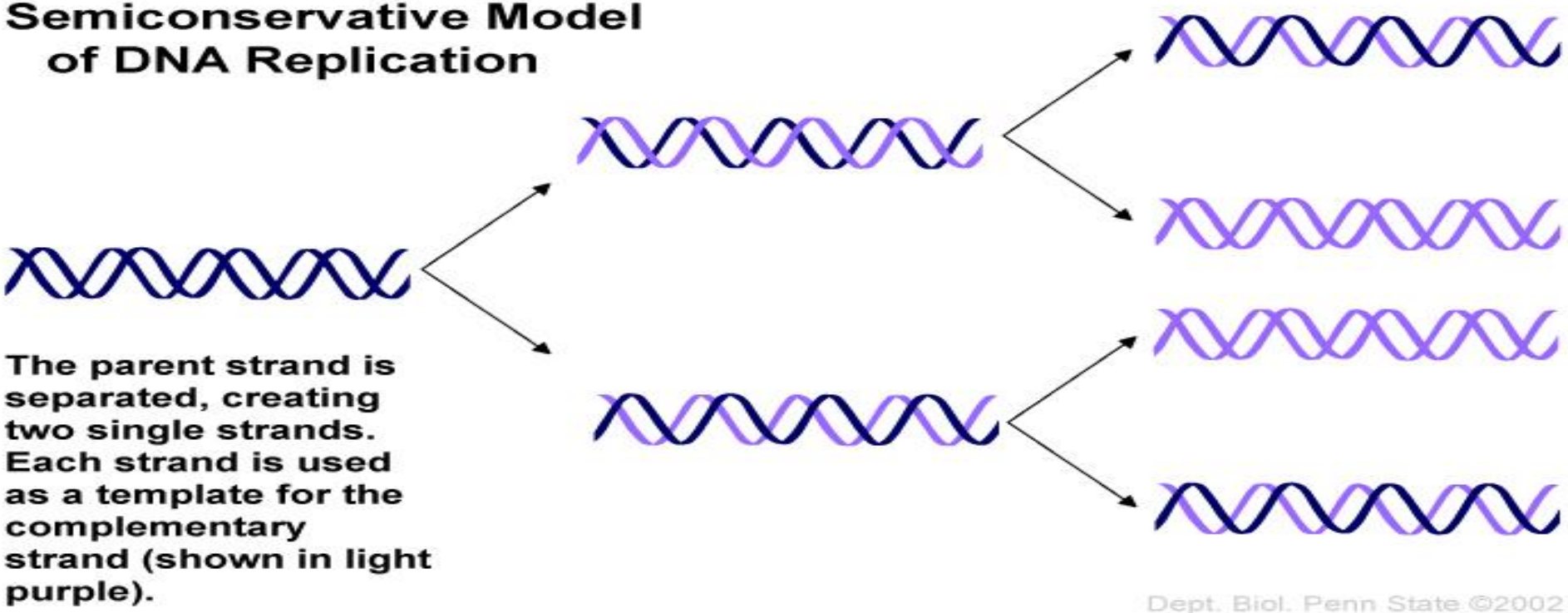





- **Base pairing allows each parental strand to serve as a template for new strand. So, new duplex is half parent template and half new DNA strand.**

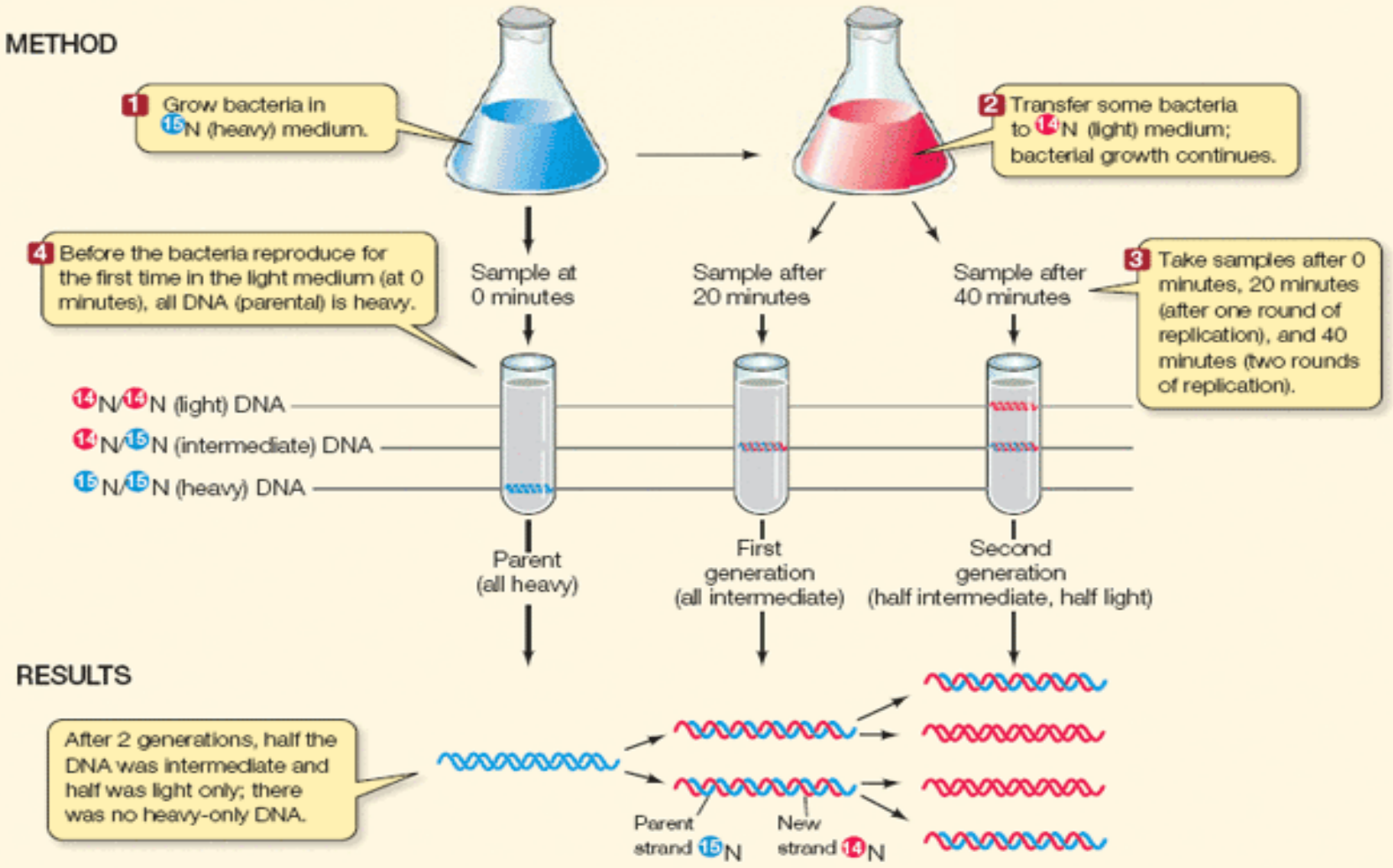
# Semiconservative Replication

Half of the parental DNA molecule is conserved in each new double helix, paired with a newly synthesized complementary strand. This is called semiconservative replication.

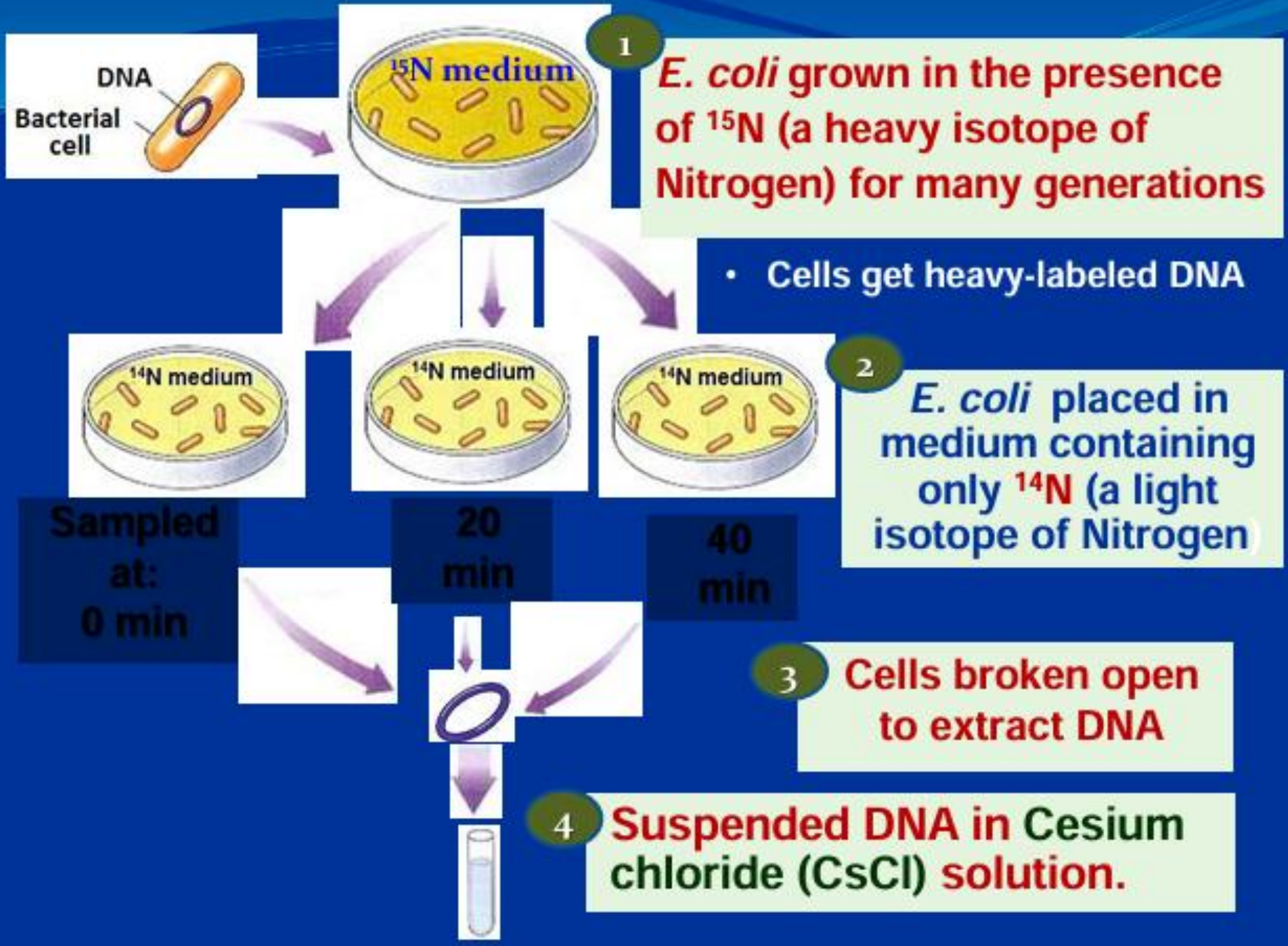


# Meselson & Stahl Experiment

**HYPOTHESIS:** DNA replicates semiconservatively.

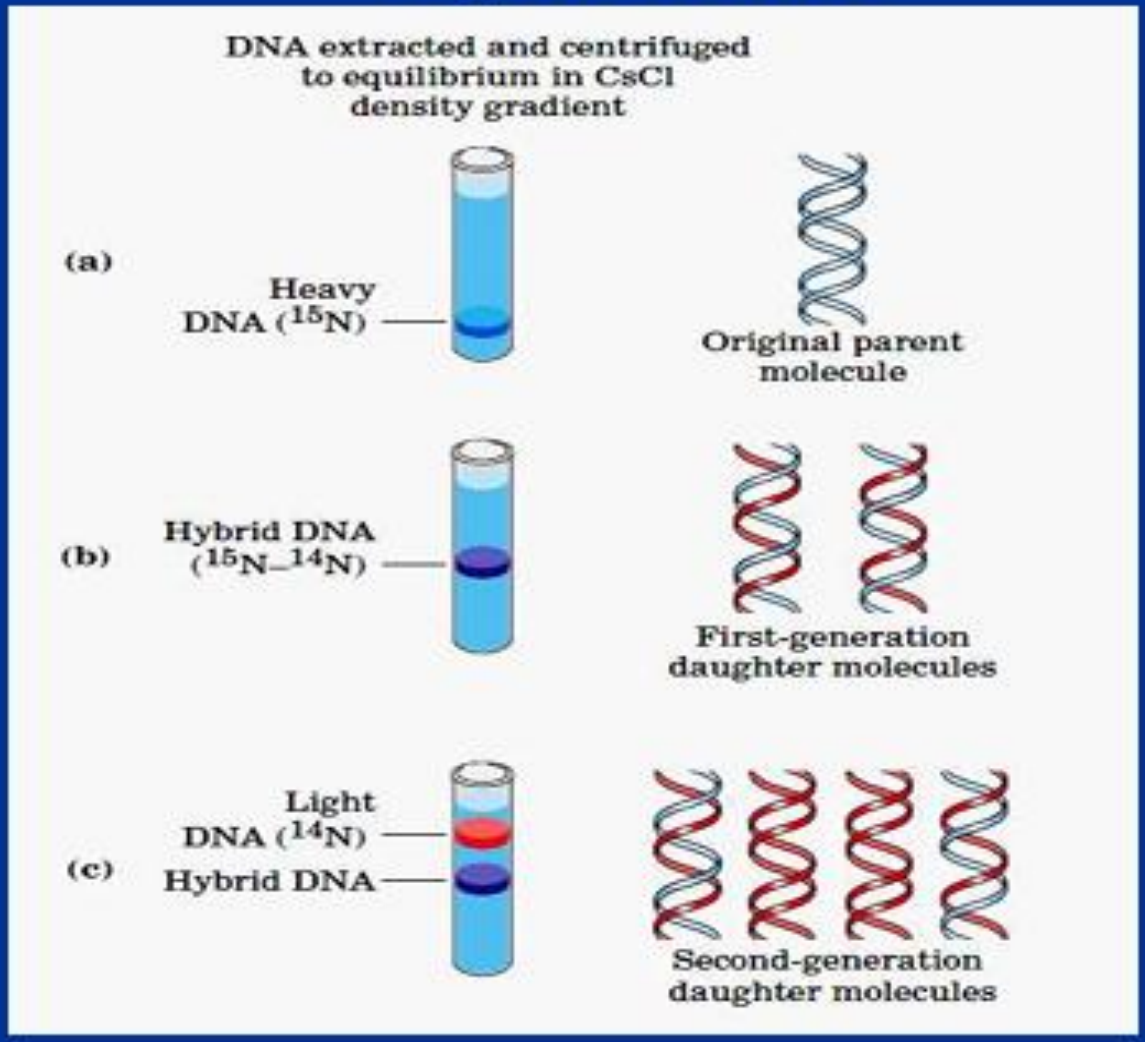



**CONCLUSION:** This pattern could only have been observed if each DNA molecule contains a template strand from the parental DNA; thus DNA replication is semiconservative.


DNA
Bacterial cell
15N medium
1
E. coli grown in the presence of 15N (a heavy isotope of Nitrogen) for many generations
• Cells get heavy-labeled DNA
14N medium
14N medium
14N medium
2
E. coli placed in medium containing only 14N (a light isotope of Nitrogen)
Sampled at: 0 min
20 min
40 min
3
Cells broken open to extract DNA
4
Suspended DNA in Cesium chloride (CsCl) solution.

# Meselson and Stahl experiment [1958]

## demonstrated Semiconservative replication



Three rounds of replication:

Original DNA

0 min

1st Round:

20 min

2nd Round:

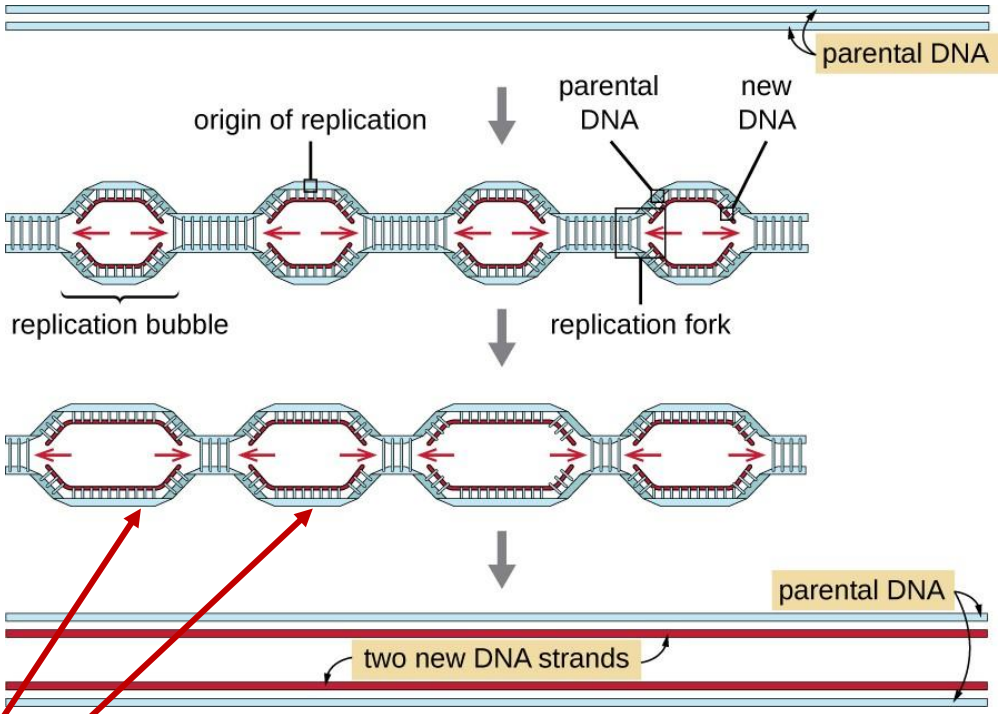
40 min

3rd Round:

60 min?

# Bidirectional replication

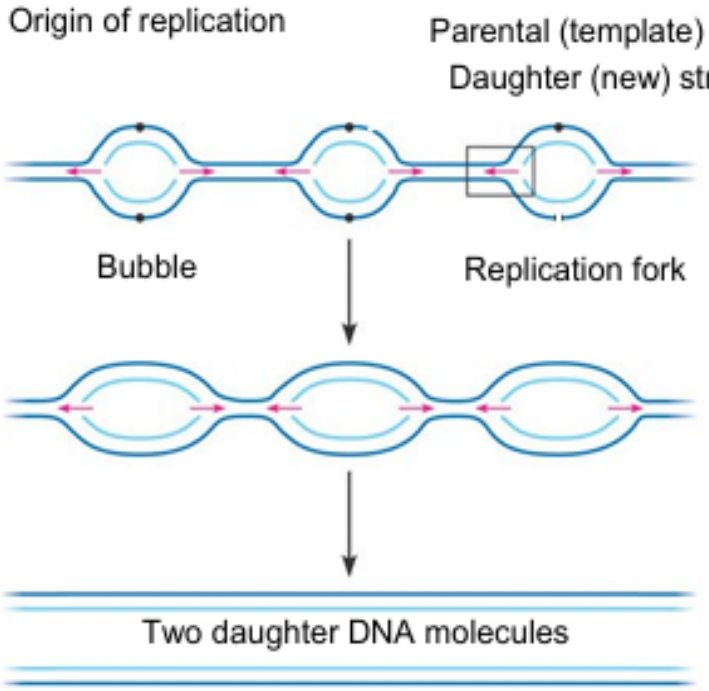
- Replication starts from unwinding the dsDNA at a particular point (called **origin / *ori* site**), followed by the synthesis on each strand.
- The parental dsDNA and two newly formed dsDNA form a Y-shape structure called **Replication fork**.




- **Many Replication bubbles are formed during bidirectional replication**

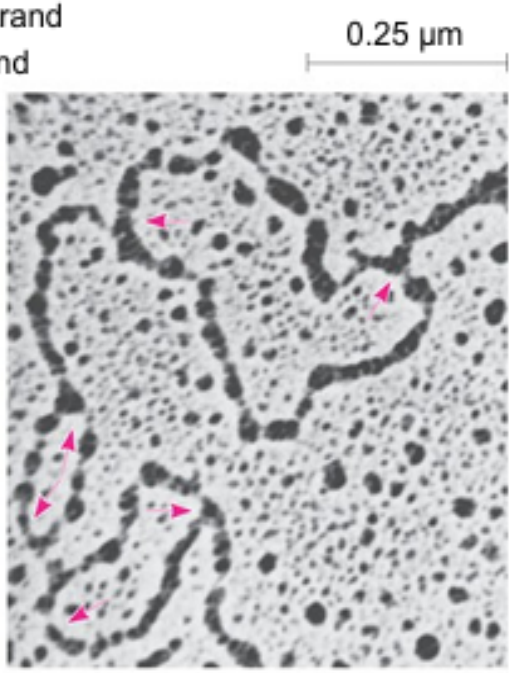
# Origins of Replication

- A eukaryotic chromosome may have hundreds or even thousands of replication origins

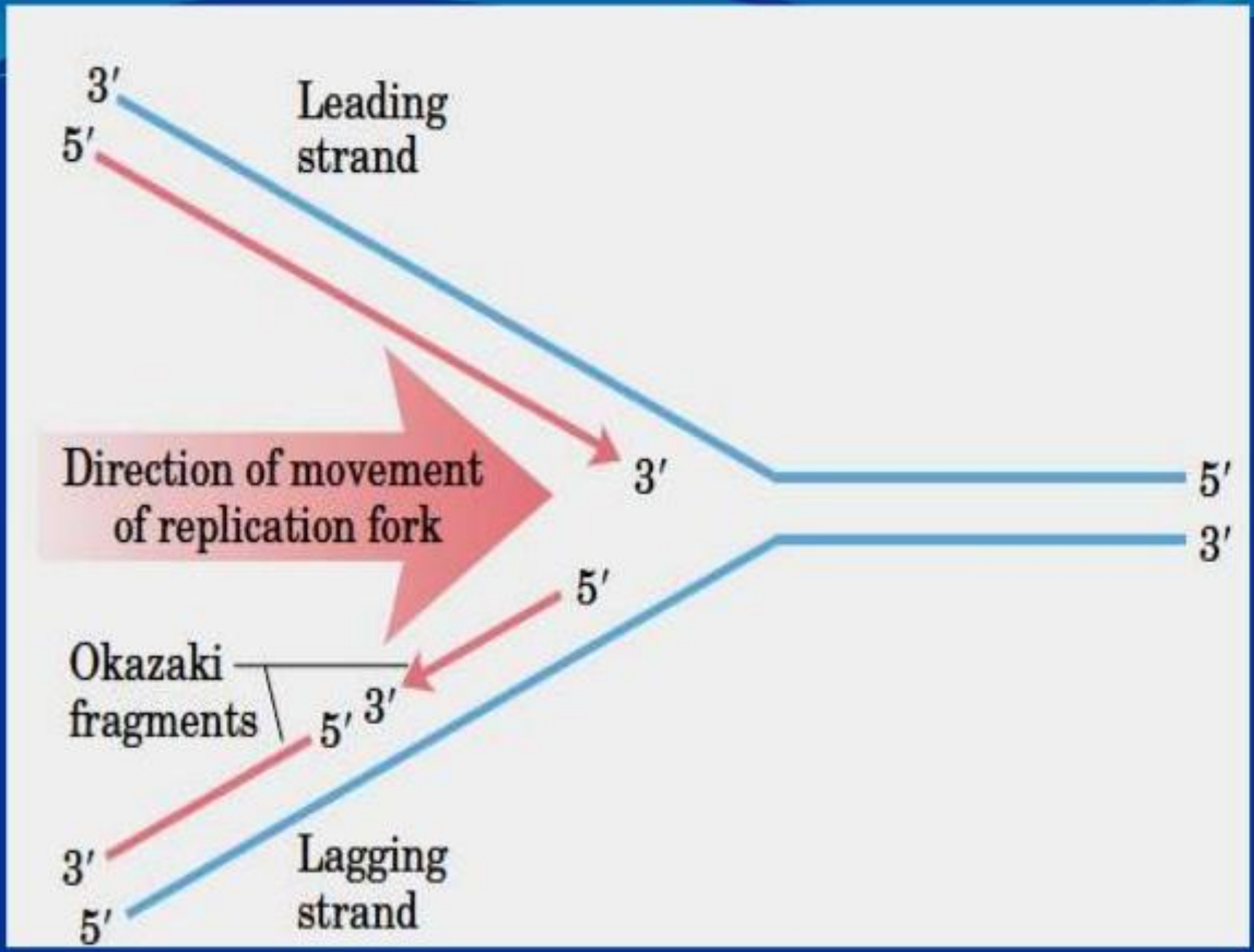



1. Replication begins at specific sites where the two parental strands separate and form replication bubbles.
2. The bubbles expand laterally, as DNA replication proceeds in both directions.
3. Eventually, the replication bubbles fuse, and synthesis of the daughter strands is complete.

**(a)** In eukaryotes, DNA replication begins at many sites along the giant DNA molecule of each chromosome.




**(b)** In this micrograph, three replication bubbles are visible along the DNA of a cultured Chinese hamster cell (TEM).


3′
5′
Leading strand
Direction of movement of replication fork
3′
5′
3′
5′
Okazaki fragments
5′
3′
3′
5′
Lagging strand

# Key points in the process of DNA replication

- **Direction of replication-** 5’ to 3’ direction
- **Leading strand-** synthesized continuously in 5’ to 3’ direction by DNA polymerase.
- **Lagging strands**-synthesized discontinuously. The short discontinuous fragments are called **“okazaki fragments”.**
- **Replication fork-** partial separation of double helix at the unzipped region of DNA is called as replication fork. Direction of fork movement is also 5’ to 3’.
- **DNA synthesis requires the four deoxynucleotide triphosphates-** dATP, dGTP, dCTP, dTTP. Nucleoside triphosphates (NTPs) have three phosphoryl groups which are attached via the **5’-hydroxyl of the 2’-deoxyribose sugar**. The innermost phosphoryl group is called alpha-phosphate, the middle one is beta & the outermost one is gamma phosphate.
- **DNA is synthesized by extending the 3’ end of the primer**. The 3’-end contains the –OH group to which new nucleotides are added. The 3’-OH group attacks the alpha phosphoryl group of incoming Nucleoside triphosphate. The leaving group is pyrophosphate which arises due to release of beta and gamma phosphates of the substrate.

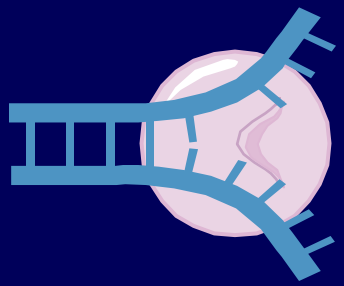
**The reaction is indicated as NTP + (NMP)n→(NMP)n+1 + PPi (pyrophosphate)**

**The breaking down of pyrophosphate by an enzyme pyrophosphatase releases energy that forms the driving force of DNA synthesis**

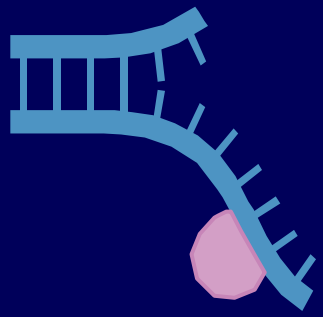
# Key enzymes that are involved in DNA replication

- **Topoisomerase (Gyrase)–** Enzyme that can break bonds and reforms the bonds. The purpose is to release the twists in DNA that are generated during DNA replication.
- **DNA helicase**: breaks the hydrogen bonds between the DNA strands. Thus, unzipping the DNA double strands
- **Single strand binding proteins** (SSBs)-keep the parental strands apart.
- **Primase-** RNA polymerase which synthesizes the primer by adding ribonucleotides that are complementary to the DNA template.
- **DNA polymerase**-synthesizes a daughter strand of DNA by adding deoxyribonucleotides.
- **DNA polymerase I in bacteria and DNA polymerase epsilon**- excises the RNA primers and fills in with DNA.
- **DNA ligase-** covalently links the Okazaki fragments. They join the fragments by phosphodiester linkage.

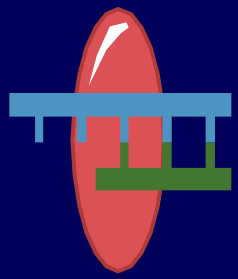
# Enzymes in DNA replication



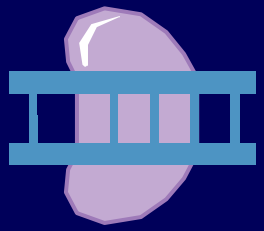
Helicase unwinds parental double helix



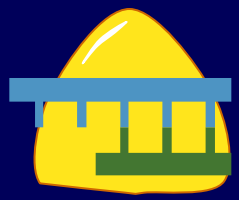
Binding proteins stabilize separate strands



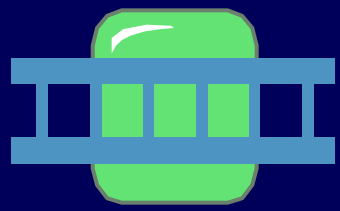
Primase adds short primer to template strand



DNA polymerase III binds nucleotides to form new strands



DNA polymerase I (Exonuclease) removes RNA primer and inserts the correct bases



Ligase joins Okazaki fragments and seals other nicks in sugar-phosphate backbone

# Steps of DNA replication

1. DNA helicase and topoisomerase unwinds the DNA double helix

↓

2. SSBs stabilizes the unwound parental strands

↓

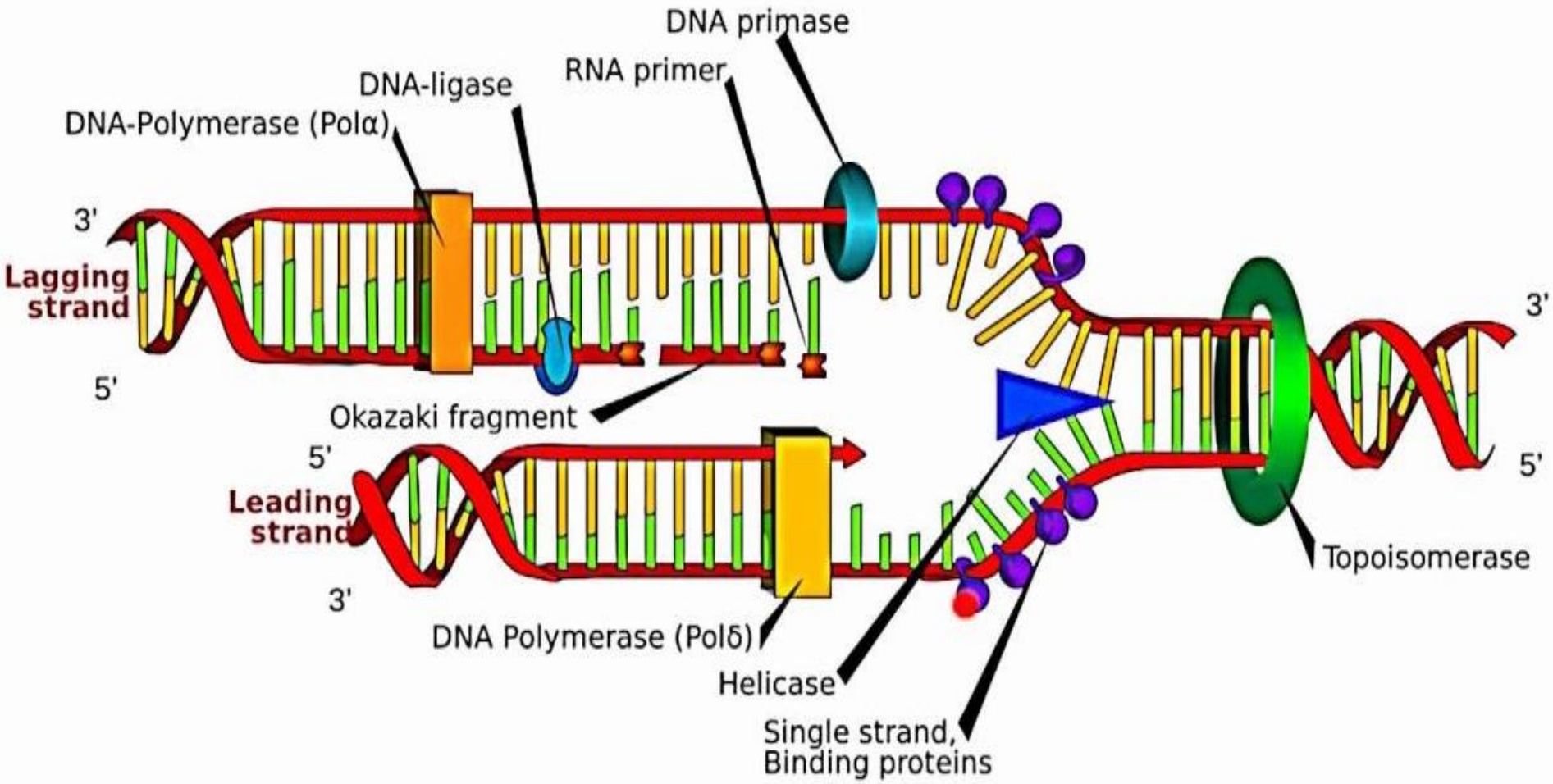
3. The leading strand is synthesized continuously in 5' to 3' direction by DNA polymerase delta.

↓

4. The lagging strand is synthesized discontinuously in 5' to 3' direction by DNA polymerase alpha. Primase synthesizes short starting fragment called as RNA primer which is extended by DNA polymerase to form Okazaki fragments.

↓

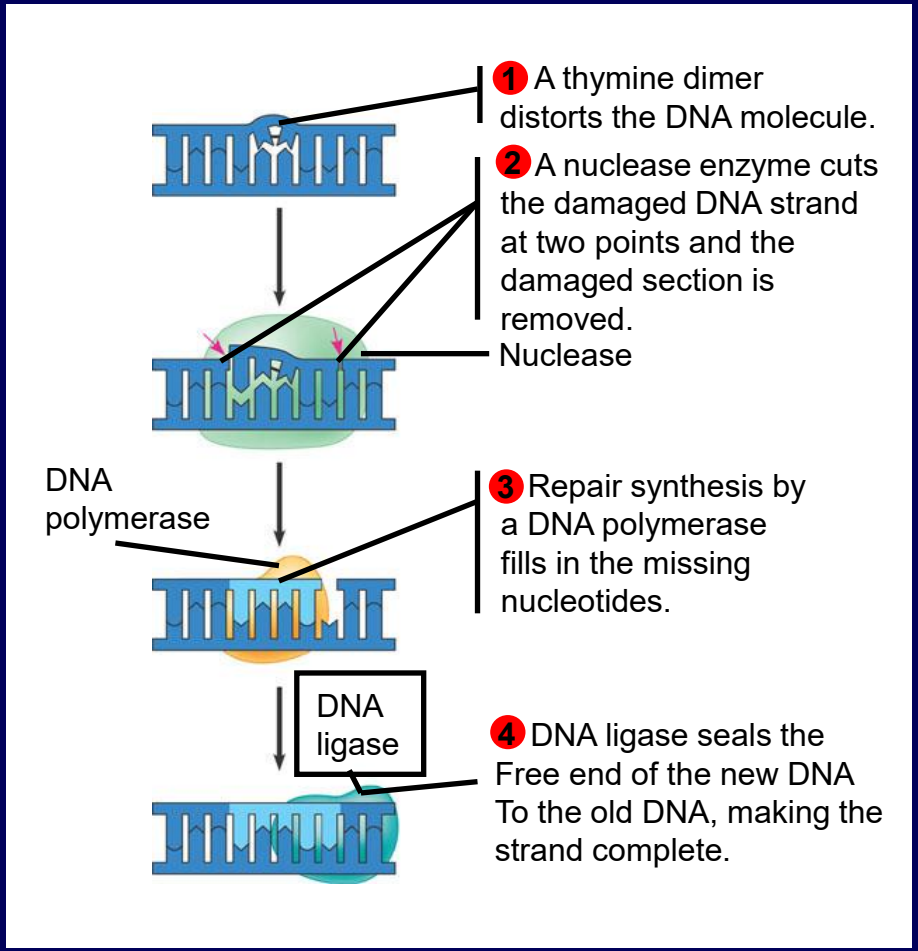
5. RNA primers are removed from the okazaki fragments by DNA polymerase epsilon. The fragments are joined by DNA ligase through phosphodiester bond




The first step in DNA replication is to separate the strands of DNA using helicase
Then DNA primase comes in to create a RNA primer
Then DNA polymerase comes in and synthesizes the new strand of DNA
Single strand binding proteins make sure the two strands don't come back together
Topoisomerase helps relieve the tension created by unwinding the DNA
Finally, DNA ligase sticks the okazaki fragments together in the lagging strand

# Proofreading and Repairing DNA

- DNA polymerases proofread newly made DNA, replacing any incorrect nucleotides
- In mismatch repair of DNA, repair enzymes correct errors in base pairing
- In nucleotide excision DNA repair nucleases cut out and replace damaged stretches of DNA



# Exonuclease functions

**5′ →3′** exonuclease activity

removes primer or excise mutated segment

**3′ →5′** exonuclease activity

excise mismatched nuleotides

